

THESIS SUMMARY: RESEARCH AND DEVELOP SOLUTIONS TO ESTIMATE ARRIVAL TIME AND EFFECTIVE ROUTING

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1 Abstract

In the era of rapid urbanization, the ability to accurately predict the Estimated Time of Arrival (ETA) has become a cornerstone of Intelligent Transportation Systems, directly impacting logistics efficiency and urban traffic management. However, providing precise estimates in megacities remains a significant challenge due to complex traffic patterns and environmental volatility. Conventional prediction models primarily rely on internal road attributes, such as segment length or historical speed limits, while often overlooking the spatial context, which includes the functional characteristics of the surrounding area that heavily influence traffic flow.

To address these limitations, this research introduces two major contributions. First, we propose a novel deep learning framework called **Point-of-Interest aware Contrastive Learning for Trajectory-based Travel Time Estimation (PoI-CL-TTE)**. The core innovation lies in integrating Points of Interest (PoI), such as schools, hospitals, and commercial centers, into open-source digital map networks to capture how different urban activities affect travel time. Furthermore, we employ a Contrastive Learning mechanism to ensure the model remains robust and accurate even when dealing with inconsistent data from community-contributed map platforms. Second, to fill a critical data gap for developing nations, we independently constructed a large-scale trajectory dataset in Ho Chi Minh City containing over one million real-world data samples. Our evaluations demonstrate that by incorporating the surrounding environment, the PoI-CL-TTE model significantly outperforms traditional methods, establishing a new and highly reliable benchmark for traffic prediction in complex urban landscapes.

2 Methodology

The PoI-CL-TTE framework is a novel architecture designed to estimate travel time by understanding urban dynamics rather than just road geometry. Its core innovation is the integration of semantic spatial contexts—specifically Points of Interest (PoIs) such as schools and commercial zones—directly into the input features. To ensure robustness against noisy crowdsourced map data, the model utilizes a Contrastive Learning mechanism powered by a Transformer Encoder. This stage acts as a powerful regularizer, forcing the model to learn invariant and highly stable structural representations of the routes.

Regarding the data flow, the PoI-enriched trajectory sequences are first processed by the Transformer to capture complex, long-range spatial-temporal dependencies. This refined global context is then fed into a Gated Recurrent Unit (GRU) to model the sequential momentum of the trip. Finally, an attention pooling layer aggregates this temporal data to output a highly precise ETA in seconds.

3 Experiments and Evaluation

Datasets Description: We evaluate the proposed model on two large-scale trajectory datasets. First, the public **Porto dataset** [1] was utilized as a global benchmark for taxi trajectory prediction. Second, we independently constructed the **Ho Chi Minh City (HCM) dataset** to address the lack of routing data in developing urban areas. This custom dataset comprises over 1.04 million trajectories collected via the TomTom API over a three-week period, specifically from November 1st to November 22nd, 2025. The data collection was scheduled daily from 5:00 to 22:00 to capture diverse traffic conditions. The samples were pre-processed to exclude short trips (under 1 km) and focus on urban routing scales typically ranging up to 15 km, ensuring the dataset accurately reflects the complex mobility patterns and high-density traffic of a megacity.

Experimental Setup and Efficiency: The training and evaluation processes were conducted on the Google Colab platform, utilizing an **NVIDIA Tesla T4 GPU (16GB VRAM)** and 16GB of system RAM. A key advantage of the PoI-CL-TTE model is its extreme architectural efficiency. While maintaining superior accuracy, the model utilizes only **1.3 million parameters**, roughly 25% of the parameter count required by previous state-of-the-art models like Mult-TTE (5.24 million parameters).

This significant reduction in complexity allows for faster training cycles and efficient deployment on resource-constrained hardware for real-time inference.

Evaluation Metrics: We utilize three standard metrics: (1) **Mean Absolute Error (MAE)** and **Root Mean Square Error (RMSE)** to measure the average magnitude of prediction errors in seconds; (2) **Mean Absolute Percentage Error (MAPE)** to evaluate the relative accuracy compared to the actual travel time, providing a clear view of the model’s precision across different trip lengths.

Table 1: Performance Comparison on Porto and HCM Datasets

Dataset	Model	MAPE (%)	RMSE (s)	MAE (s)
Porto	Linear Regression	25.15	213.34	145.99
	DeepTTE [2]	16.28	156.05	103.53
	MulT-TTE [3]	16.73	136.12	92.31
	PoI-CL-TTE (Ours)	14.88	128.78	85.37
HCM City	Linear Regression	14.18	261.81	198.58
	DeepTTE [2]	6.30	105.07	79.75
	MulT-TTE [3]	4.40	76.60	53.93
	PoI-CL-TTE (Ours)	1.91	37.77	25.59

Result Analysis: PoI-CL-TTE consistently outperforms baselines across all metrics. On the HCM dataset, it nearly halves absolute errors (RMSE: **37.77s**, MAE: **25.59s**) and achieves a remarkably low MAPE of **1.91%**, a **56.5% improvement** over the state-of-the-art MulT-TTE [3]. This confirms that integrating PoI-based spatial context effectively mitigates the volatility of dense urban traffic.

4 Conclusion and Contributions

This research provides a comprehensive solution for Estimated Time of Arrival in complex urban environments. The project’s contributions are summarized as follows:

- We successfully developed the PoI-CL-TTE framework, which is the first to combine Point-of-Interest semantics with Contrastive Learning for travel time estimation. This allows for a deeper understanding of urban dynamics and increased robustness against map data noise.
- We constructed a large-scale, real-world traffic benchmark for Ho Chi Minh City. This fills a significant data gap for developing megacities and provides a vital resource for future Intelligent Transportation Systems research.
- The project achieves high accuracy with significant architectural optimization, enabling real-time navigation services on resource-constrained devices, thus laying a robust foundation for smarter urban logistics.

References

- [1] Kaggle, “ECML/PKDD 15: Taxi trajectory prediction (I),” <https://www.kaggle.com/c/pkdd-15-predict-taxi-service-trajectory-i>, 2015.
- [2] D. Wang, J. Zhang, W. Cao, J. Li, and Y. Zheng, “When will you arrive? Estimating travel time based on deep neural networks,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018.
- [3] T. Liao, L. Han, Y. Xu, T. Zhu, L. Sun, and B. Du, “Multi-faceted route representation learning for travel time estimation,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 9, pp. 11782–11793, 2024.